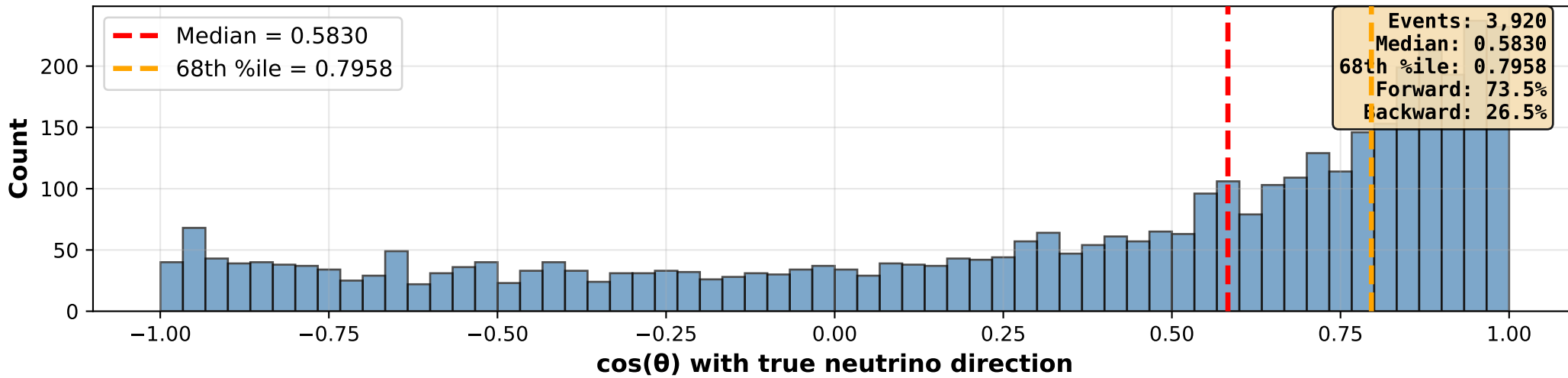


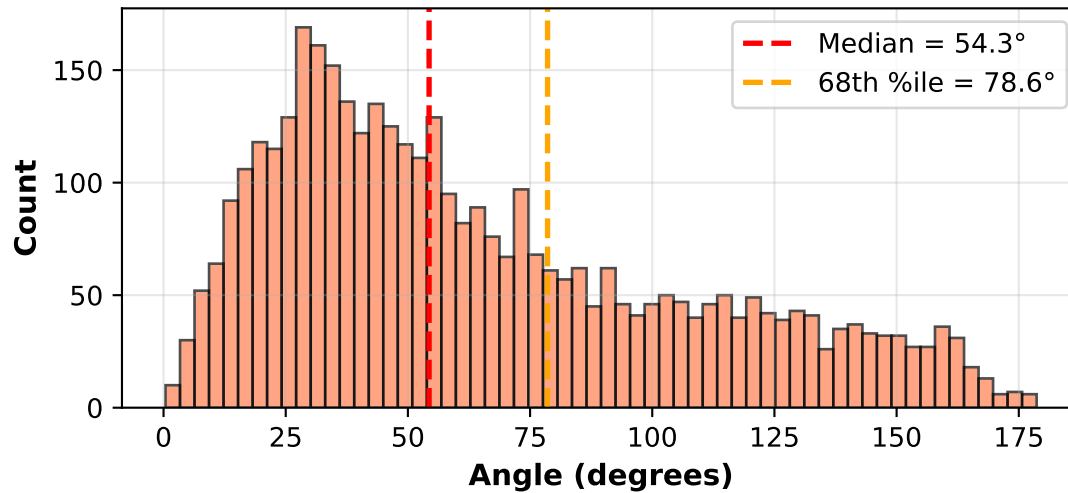
ED Model + MCMC Pointing Performance

Overall Results (98 categories, 3,920 events)

Overall $\cos(\theta)$ Distribution



Angle Distribution



SUMMARY STATISTICS

Total Events: 3,920
Successful Categories: 98/100

Cosine Distribution:
Median: 0.5830
Mean: 0.3541
68th percentile: 0.7958
Std deviation: 0.5989

Angle Distribution:
Median: 54.3°
Mean: 65.4°
68th percentile: 78.6°

Pointing Direction:
Forward ($\cos > 0$): 2,883 (73.5%)
Backward ($\cos < 0$): 1,037 (26.5%)

COMPARISON TO BASELINES

Best possible (true e^- dirs):
 $\cos(\theta)$ 68% = 0.9980

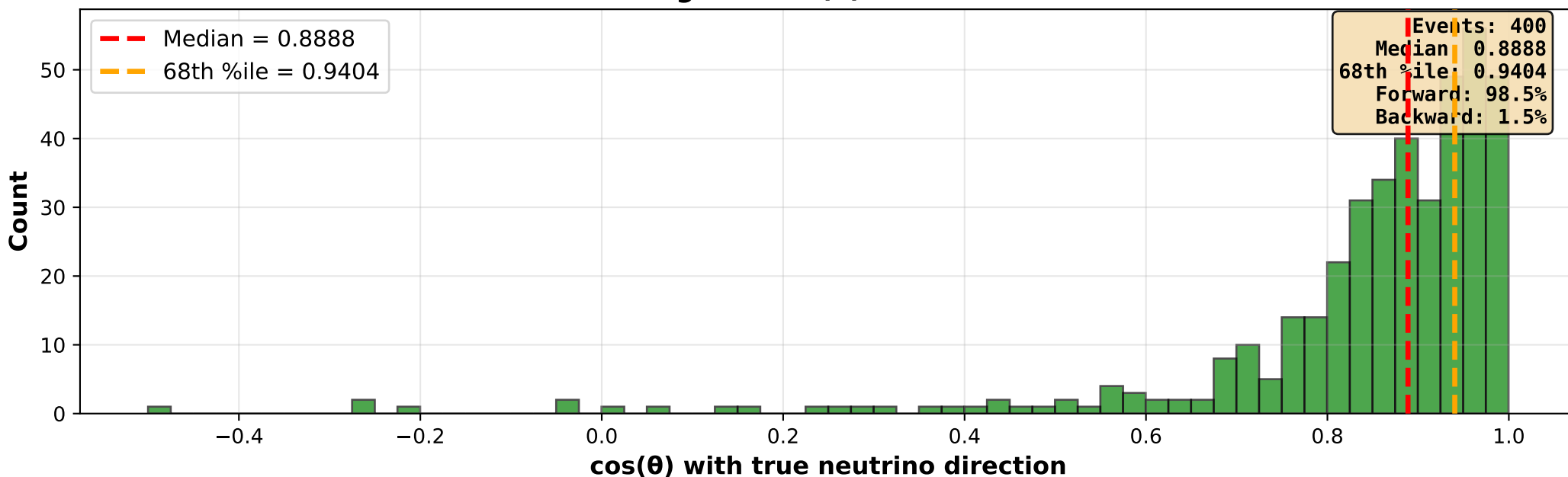
Current (ED model):

ED Model + MCMC Pointing Performance

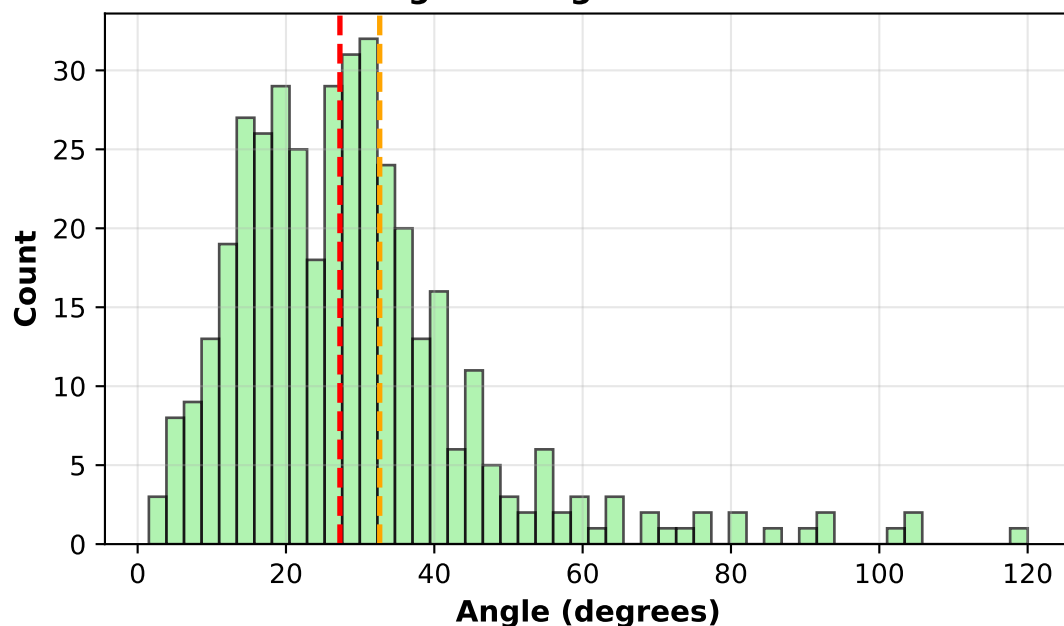
Z-Aligned Neutrinos (within $\pm 30^\circ$ of Z-axis)

10 categories, 400 events

Z-Aligned: $\cos(\theta)$ Distribution



Z-Aligned: Angle Distribution



Z-ALIGNED CATEGORIES (within $\pm 30^\circ$ of Z-axis)

Categories: 10
Events: 400
 $\cos(\theta)$ 68%: 0.9404
Median angle: 27.3°
Forward: 98.5%

Best categories:

cat000094: 0.9685
cat000035: 0.9680
cat000060: 0.9564
cat000011: 0.9511
cat000007: 0.9417

Worst categories:

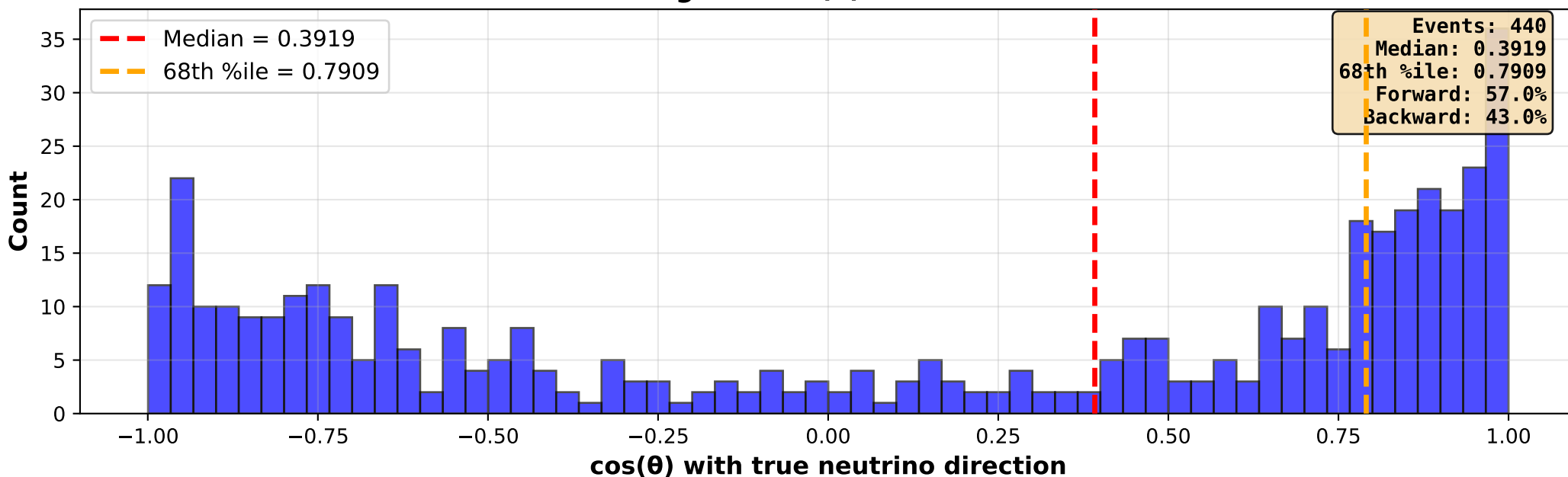
cat000036: 0.8888
cat000026: 0.8868
cat000062: 0.8795

ED Model + MCMC Pointing Performance

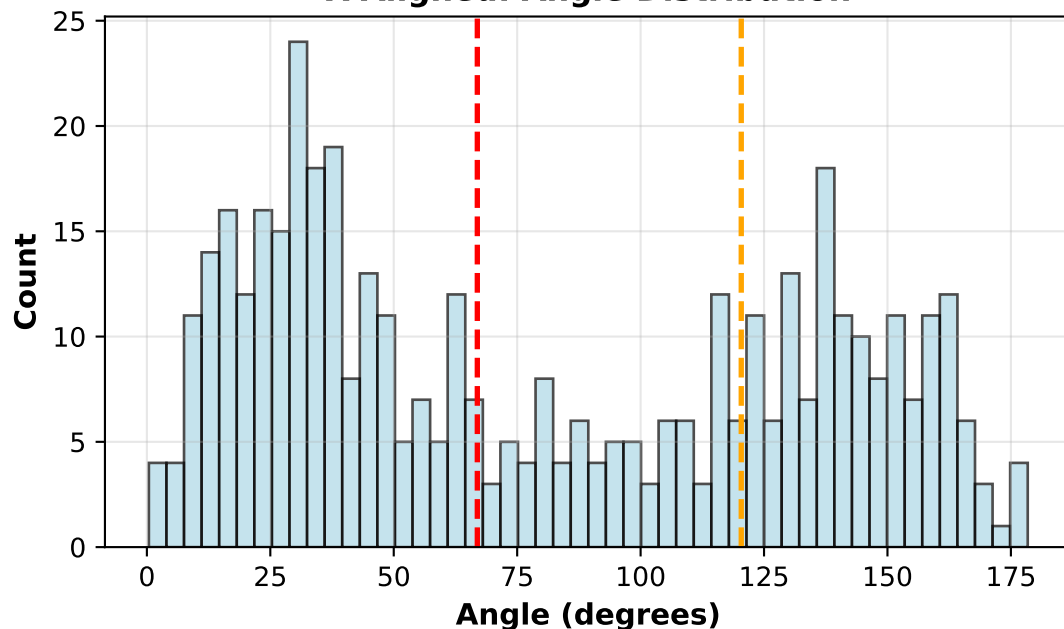
X-Aligned Neutrinos (within $\pm 30^\circ$ of X-axis)

11 categories, 440 events

X-Aligned: $\cos(\theta)$ Distribution



X-Aligned: Angle Distribution



X-ALIGNED CATEGORIES (within $\pm 30^\circ$ of X-axis)

Categories: 11
Events: 440
 $\cos(\theta)$ 68%: 0.7909
Median angle: 66.9°
Forward: 57.0%

Best categories:

cat000039: 0.9102
cat000042: 0.9022
cat000021: 0.8967
cat000029: 0.7930
cat000038: 0.7837

Worst categories:

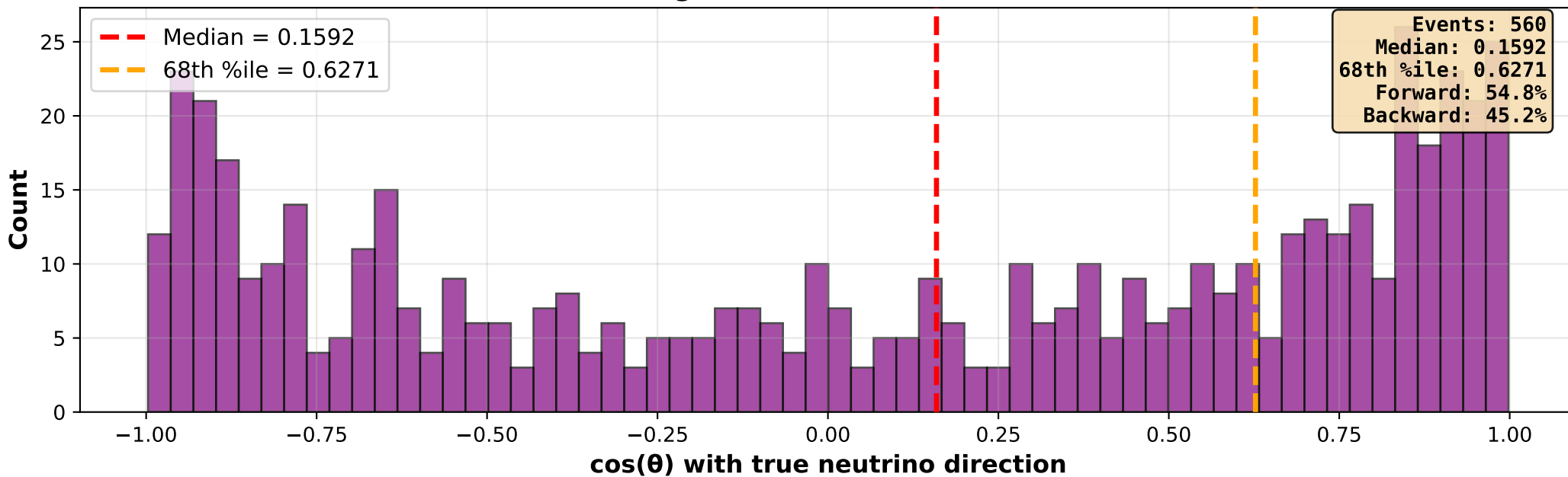
cat000003: -0.1246
cat000078: -0.3369
cat000103: -0.6432

ED Model + MCMC Pointing Performance

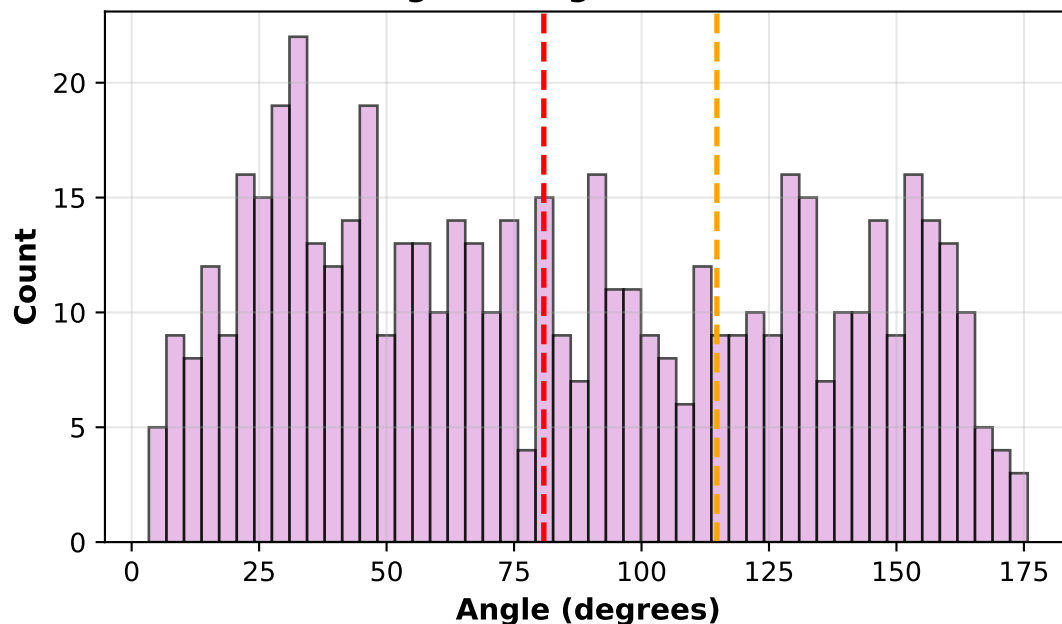
Y-Aligned Neutrinos (within $\pm 30^\circ$ of Y-axis)

14 categories, 560 events

Y-Aligned: $\cos(\theta)$ Distribution



Y-Aligned: Angle Distribution



Y-ALIGNED CATEGORIES (within $\pm 30^\circ$ of Y-axis)

Categories: 14
Events: 560
 $\cos(\theta)$ 68%: 0.6271
Median angle: 80.8°
Forward: 54.8%

Best categories:

cat000086: 0.8660
cat000075: 0.8647
cat000091: 0.8613
cat000096: 0.7959
cat000041: 0.6906

Worst categories:

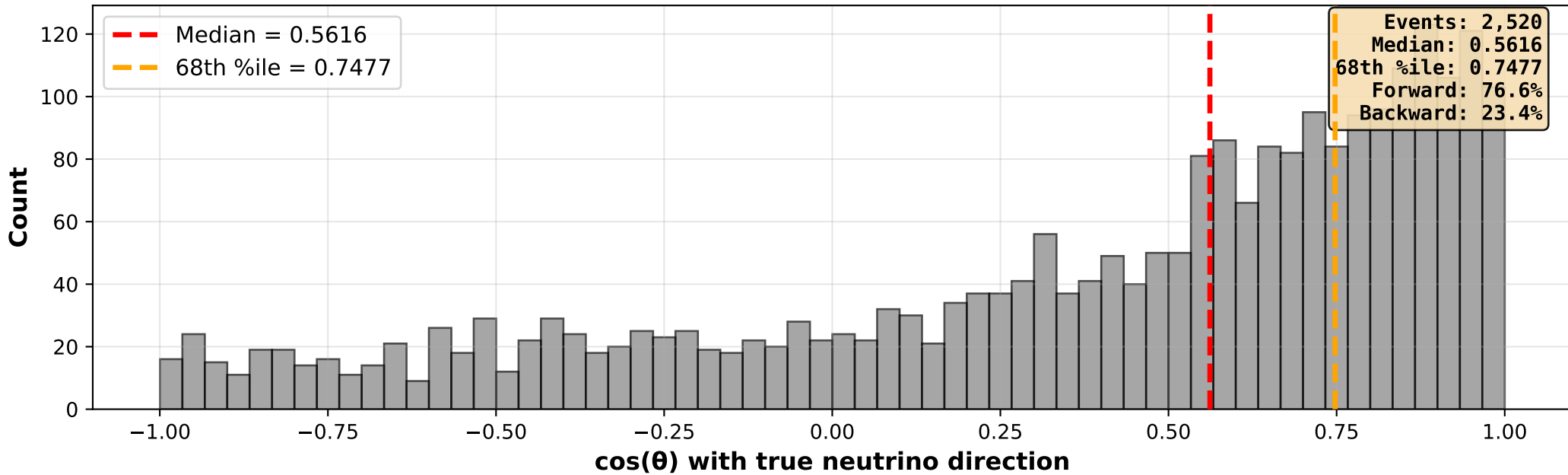
cat000025: 0.0819
cat000013: 0.0322
cat000046: 0.0056

ED Model + MCMC Pointing Performance

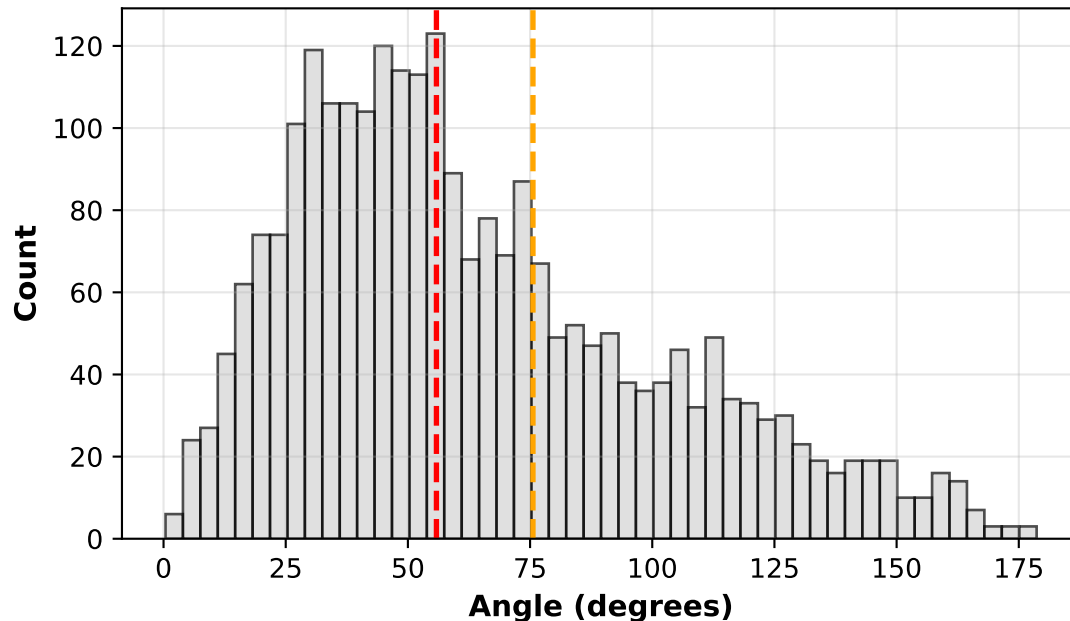
Other Orientations (not aligned with any axis)

63 categories, 2,520 events

Other Orientations: $\cos(\theta)$ Distribution



Other: Angle Distribution



OTHER ORIENTATIONS (not within $\pm 30^\circ$ of any axis)

Categories: 63
Events: 2,520
cos(θ) 68%: 0.7477
Median angle: 55.8°
Forward: 76.6%

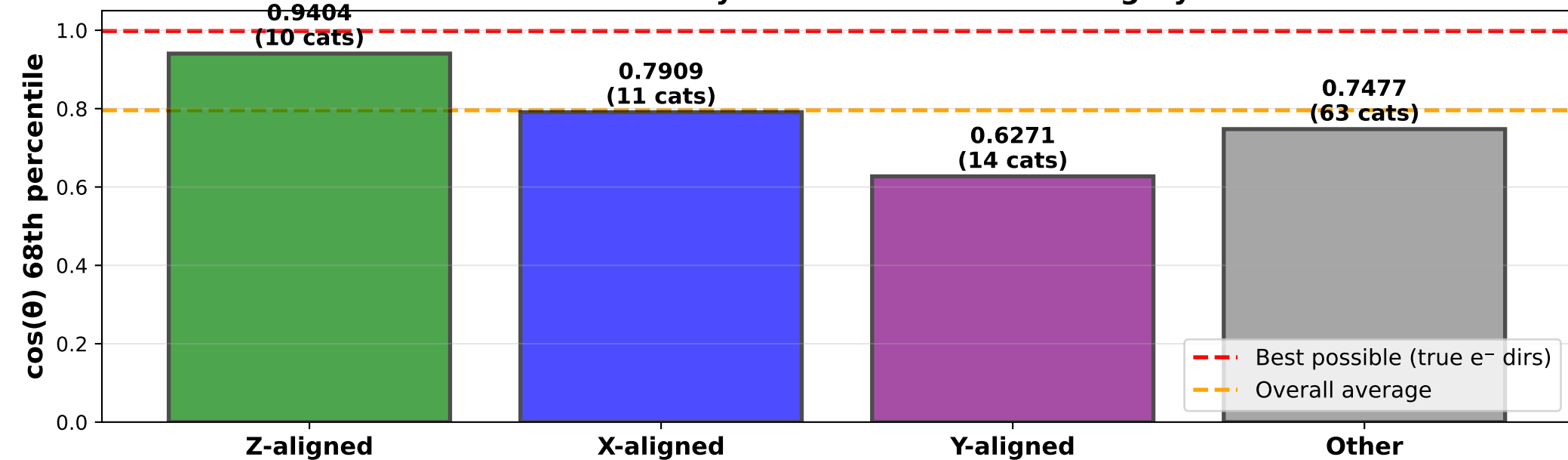
Best categories:
cat000080: 0.9668
cat000065: 0.9427
cat000027: 0.9173
cat000088: 0.9095
cat000085: 0.9079

Worst categories:
cat000040: 0.1805
cat000032: 0.0779
cat000010: -0.3100

ED Model + MCMC Pointing Performance

Performance Comparison by Neutrino Direction

Performance by Neutrino Direction Category



PERFORMANCE SUMMARY BY DIRECTION

Category	N_cats	N_events	$\cos(\theta)$ 68%	Median°	Forward%
Z-aligned	10	400	0.9404	27.3°	98.5%
X-aligned	11	440	0.7909	66.9°	57.0%
Y-aligned	14	560	0.6271	80.8°	54.8%
Other	63	2,520	0.7477	55.8°	76.6%
Overall	98	3,920	0.7958	54.3°	73.5%

KEY FINDINGS

- Z-aligned neutrinos perform BEST ($\cos$ 68% = 0.9404)
- Other orientations perform WORST ($\cos$ 68% = 0.7477)
- Overall performance is dominated by detector geometry:
 - Parallel to wire planes (Z): Excellent reconstruction
 - Perpendicular to wire planes: Poor reconstruction
- Best possible (true e^- directions): $\cos$ 68% = 0.9980
- Current gap: $\Delta\cos$ = 0.2022